

CAR-HAL: Attention-Based Hallucination Detection in RAG

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Abstract. Retrieval-Augmented Generation (RAG) systems remain vulnerable to hallucination even when relevant context is retrieved. Existing detection methods relying on surface-level lexical overlap and semantic similarity can be fooled by coincidental vocabulary overlap—a failure mode we term *Spurious Faithfulness* (SF). We introduce **CAR-HAL**, an attention-aware framework built around the *Context Attention Ratio* (CAR), which measures the proportion of generative attention directed at retrieved context tokens across all transformer layers. CAR is combined with an *Attention-Output Consistency Score* (AOCS), an *Intra-Context Contradiction Score* (ICCS), and a fine-tuned DeBERTa-v3-base classifier in a logistic regression ensemble. CAR-HAL achieves 92.5% accuracy and 92.0% macro-F1 on the RGB benchmark, outperforming all baselines ($p < 0.05$). Spurious Faithfulness affects 53–65% of hallucinated samples across our evaluation settings, meaning surface-only detectors miss more than half of all generation-tier hallucinations. We further identify a precise prompt-topology boundary that determines when attention-based detection succeeds, yielding mechanistic insight into the scope and limits of the approach.

Keywords: Hallucination detection, Retrieval-augmented generation, Attention mechanisms, Context attention ratio, Spurious faithfulness, Large language models

1 Introduction

Large language models (LLMs) now underpin most question-answering systems, predominantly through the Retrieval-Augmented Generation (RAG) paradigm [11,12], which conditions generation on dynamically retrieved documents to ground responses in verifiable knowledge. Hallucination nonetheless persists even when the relevant document is retrieved successfully—the model simply ignores it and falls back on parametric memory. This generation-tier failure is distinct from retrieval-tier failure and demands detection methods that probe model-internal behaviour rather than comparing output surfaces [11,12].

In high-stakes deployments—medical QA, legal analysis, enterprise knowledge management—a fluent, topically relevant response that is factually divorced from its retrieved context can cause direct harm, since a user has little reason

to question an answer that reads as confident and well-sourced. The problem is compounded because hallucinated outputs often *appear* faithful: they reuse vocabulary from the retrieved passage, score highly on lexical overlap, and pass entailment filters. We term this *Spurious Faithfulness* (SF) and show empirically that it affects 53–65% of hallucinated samples across diverse evaluation settings—meaning a sizeable share of hallucinations are effectively invisible to the detectors most commonly deployed in production RAG pipelines today.

Existing detection approaches fall into two families. *Post-hoc surface methods* such as ROUGE and BERTScore compare generated answers to context at the token or embedding level [5], and are cheap but blind to the generative mechanism. *Entailment-based methods*, including LLM-as-a-judge frameworks such as FaithJudge [7] and metamorphic testers such as MetaRAG [8], assess logical faithfulness more robustly but remain silent about *why* a hallucination arose. Neither family detects the case where a model reproduces surface vocabulary from context while actually reasoning from parametric memory, and both typically require an auxiliary LLM call per sample, adding cost and latency of their own.

We argue that transformer attention weights are a direct, interpretable, mechanistically grounded signal for this problem: a hallucinating model must allocate less causal attention to retrieved context tokens during generation. CAR quantifies exactly this, building on mechanistic interpretability work showing that attention patterns reflect knowledge-source selection [2,10].

This paper makes four contributions. (1) **CAR and AOCS**: a forward-pass attention framework quantifying context reliance and answer-context semantic coherence without gradient computation. (2) **Empirical characterisation of Spurious Faithfulness**: the first systematic measurement of how often surface metrics incorrectly certify hallucinated outputs as faithful. (3) **Mechanistic domain-boundary analysis**: a quantitative account of the prompt-topology conditions under which attention-based signals succeed and fail. (4) **A hallucination taxonomy** grounded in attention dynamics, categorising failures into parametric, grounding_conflict, and spurious_faithful classes.

2 Related Work

Work most relevant to CAR-HAL spans three threads: taxonomic frameworks characterising what hallucination is, detection methods operating on output surfaces, and internal-signal approaches that probe model-internal representations.

2.1 Hallucination Taxonomies and the RAG Failure Landscape

Ji *et al.* [12] established the foundational taxonomy, distinguishing *intrinsic* hallucinations—outputs contradicting the source—from *extrinsic* ones, which introduce information the source can neither confirm nor refute. This decoupled faithfulness from factual accuracy: a model can be unfaithful to a document it was given without producing anything false about the world, and vice

versa. Zhang and Zhang [11] extended it to RAG, separating retrieval-tier failures (the relevant document is never retrieved) from generation-tier failures (it is retrieved but the model diverges from it anyway)—a distinction that matters because the two demand different fixes: better retrieval versus detection methods that probe what the model does *after* receiving context. CAR-HAL addresses exclusively the generation-tier case, assuming retrieval has succeeded. Spurious Faithfulness—hallucinated outputs that coincidentally reproduce surface vocabulary and thereby evade lexical/embedding detection—is a sub-category of generation-tier failure that existing taxonomies do not explicitly name; its systematic characterisation is a contribution of this work.

2.2 Surface-Level and Entailment-Based Detection

ROUGE/BLEU-style n-gram overlap and BERTScore’s embedding similarity remain the most widely deployed detectors, largely because they require no additional model calls beyond the generation itself, but both are blind to the generative mechanism: a model that ignores retrieved context and generates from parametric memory still scores highly if the output happens to share vocabulary with the passage—precisely the SF failure mode. RAGAS [5] extends surface evaluation to the pipeline level but inherits the same overlap-based blindness. Entailment-based methods are more robust: FaithJudge [7] uses an LLM-as-a-judge to check whether each claim traces back to the retrieved passage, while MetaRAG [8] uses metamorphic testing, checking answer consistency across paraphrased queries. Both resist coincidental vocabulary overlap better than surface metrics, but neither explains *why* a hallucination arose—they treat the model as a black box, leaving open whether it failed to attend to context, attended but favoured parametric memory, or faced genuine ambiguity in the context itself. CAR-HAL answers exactly these questions by examining the generative process from within.

2.3 Internal-Signal Detection Methods

Chen *et al.* [10] showed hallucinated and faithful generations are linearly separable in hidden-state activation space, establishing that internal representations carry hallucination signals absent from the output surface—the mechanistic grounding for attention-based approaches. ReDeEP [2] operationalises this via causal tracing, decomposing each token into parametric versus retrieved-knowledge contributions through targeted interventions; it achieves fine-grained attribution but requires a separate intervention per sequence. LRP4RAG [3] instead uses Layer-wise Relevance Propagation, backpropagating relevance from output to input tokens, at the cost of a full backward pass per sequence. CAR operationalises the same internal/external distinction through raw attention weights from a single additional forward pass, achieving comparable interpretability at substantially lower cost—though attention is a noisier causal proxy, so CAR’s effectiveness depends on structural conditions, particularly the context-to-total token ratio r_{ctx} .

LUMINA [4] instead decomposes retrieved context itself into reliable/unreliable segments, targeting context errors the model faithfully reproduces—a failure mode that originates in the context rather than in how the model attends to it. SINDEX [6] measures distributional inconsistency between outputs and reference distributions, treating hallucination as a statistical outlier rather than as context-output divergence. Neither targets the generation-tier grounding failure CAR-HAL addresses, but both identify complementary failure modes that a comprehensive detection system would eventually need to cover alongside it. CAR-HAL’s taxonomy is intended to provide a shared vocabulary relating these approaches and clarifying where each one’s scope of applicability begins and ends.

3 Methodology

CAR-HAL processes each (query, context, answer) triple through four sequential phases (Fig. 1).

Phase 1 – Hybrid Retrieval. FAISS dense retrieval (intfloat/e5-base-v2, 768-d) combines with BM25 sparse retrieval. Top-50 candidates from each retriever are union-merged and reranked:

$$\text{score}(d) = \alpha \cdot s_{\text{faiss}}(d) + (1 - \alpha) \cdot s_{\text{bm25}}(d), \quad \alpha = 0.6. \quad (1)$$

A retrieval gate short-circuits queries below τ_{ret} , labelling them `retriever_failure` before downstream processing.

Phase 2 – Answer Generation. Mistral-7B-Instruct-v0.2 (4-bit NF4 quantisation, single 16 GB GPU) generates answers via greedy decoding (`do_sample=False`, `max_new_tokens=80`). ICCS runs in parallel on the retrieved context, adding no sequential latency.

Phase 3 – Post-Hoc Attention Analysis. After generation, a single additional forward pass with `output_attentions=True` yields tensors $\mathbf{A}_l \in \mathbb{R}^{H \times T \times T}$ for all $L=32$ layers, from which we extract layer-mean $\overline{\text{CAR}}$, layer drift $\Delta\text{CAR} = \text{CAR}_{\text{late}} - \text{CAR}_{\text{early}}$, and inter-head variance $\bar{\sigma}_{\text{head}}$. Running this as a separate forward pass rather than caching attentions during generation keeps the generation loop itself untouched, so the analysis can be applied retroactively to logs from an existing deployment without modifying the serving pipeline. A two-component GMM fitted on hallucinated training CAR values sets the CAR_{LOW} threshold.

Phase 4 – Logistic Regression Detection. All 12 signals (Eq. 6) feed a class-balanced L2-regularised logistic regression ($C=1.0$) trained on 1,980 pooled samples (180 RGB calibration + 1,800 RAGTruth), adding ≈ 340 ms per sample on a T4 GPU.

3.1 Context Attention Ratio (CAR)

CAR measures the fraction of total causal attention directed at retrieved context tokens during generation. For attention tensor $\mathbf{A}_l$ at layer l , the head-averaged

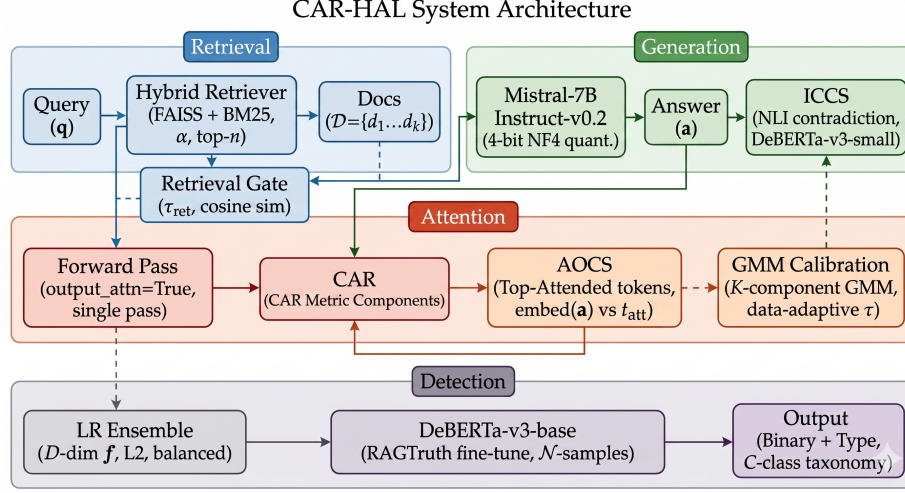


Fig. 1: CAR-HAL system architecture. The post-hoc attention analysis and logistic regression ensemble are the novel components.

ratio at generation position $t > t_{\text{prompt}}$ is:

$$\text{CAR}_l(t) = \frac{\sum_{h=1}^H \sum_{j=c_s}^{c_e} A_l^h(t, j)}{\sum_{h=1}^H \sum_{j=0}^t A_l^h(t, j)}, \quad (2)$$

where c_s and c_e are context span boundaries. The layer-mean is:

$$\overline{\text{CAR}} = \frac{1}{L} \sum_{l=1}^L \text{mean}_t [\text{CAR}_l(t)]. \quad (3)$$

A negative layer drift ΔCAR indicates parametric-memory-dominant generation [10].

3.2 Attention-Output Consistency Score (AOCS)

AOCS measures semantic coherence between what the model attended to and what it generated. The top-50th-percentile attended tokens decode into a span t_{attended} , and:

$$\text{AOCS} = \cos(\text{embed}(a), \text{embed}(t_{\text{attended}})). \quad (4)$$

CAR and AOCS jointly define a four-quadrant grouping, derived from their score combinations rather than independent expert categorisation: high CAR + high AOCS = faithful; low CAR + low AOCS = parametric hallucination; high CAR + low AOCS = **grounding_conflict**; low CAR + high AOCS = Spurious Faithfulness. We use “taxonomy” throughout for convenience, while noting in Section 6 that its correspondence to genuinely distinct phenomena has not been validated against human annotation.

Table 1: Logistic regression feature coefficients (positive = hallucination; negative = faithfulness).

Feature	Coeff.	Direction
CAR_{late}	+0.9579	hallucination
$\overline{CAR}$	−0.8168	faithfulness
Token Overlap	+0.3267	hallucination
ΔCAR	−0.3039	faithfulness
AOCS	−0.2500	faithfulness
p_{s1} (DeBERTa prob.)	+0.2245	hallucination
Semantic Similarity	+0.1720	hallucination
BERTScore-F1	−0.1470	faithfulness
b_{s1} (DeBERTa binary)	+0.0309	hallucination
$ICCS_{norm}$	−0.0119	faithfulness
$\bar{\sigma}_{head}$	−0.0019	faithfulness
<code>span_ok</code>	+0.0000	—

3.3 Intra-Context Contradiction Score (ICCS)

ICCS quantifies internal inconsistency within retrieved context. Up to 10 sentences are sampled and all pairs evaluated via bidirectional NLI (`cross-encoder/nli-deberta-v3-small`):

$$ICCS = \frac{|\{(s_i, s_j) : NLI(s_i, s_j) = \text{CONTRADICTION}\}|}{\binom{n}{2}}. \quad (5)$$

A per-document baseline $\delta_{base} = 0.05(n_{docs} - 1)$ accounts for natural cross-document contradiction rates.

3.4 Logistic Regression Ensemble

The 12-dimensional feature vector is:

$$\mathbf{f} = [\overline{CAR}, \Delta CAR, CAR_{late}, BERTScore, TokOvlp, SemSim, ICCS_{norm}, p_{s1}, b_{s1}, AOCS, \text{span_ok}, \bar{\sigma}_{head}] \quad (6)$$

Table 1 reports learned coefficients. CAR_{late} (+0.9579) is the strongest hallucination indicator, confirming late-layer attention as the primary site of parametric vs. context-grounded divergence. Token Overlap carries a *positive* hallucination coefficient (+0.3267)—the direct signature of Spurious Faithfulness.

4 Experimental Setup

4.1 Benchmarks and Data Splits

We evaluate on the **RGB benchmark** [9] (500 items across three testbeds): *noise* (300, adversarial documents), *counterfactual* (100, documents contradicting common knowledge), and *integration* (100, multi-hop synthesis). A 60/40

Table 2: Held-out RGB results ($n=120$ overall; $n=40$ per testbed). Per-testbed CIs are $\pm 31\text{pp}$; directional only. Best in **bold**.

Method	Acc.	Noise	Cntft.	Integ.	Macro-F1
Token Overlap (B1)	46.7%	77.5%	85.0%	0.0%	31.8%
BERTScore (B2)	67.5%	100.0%	67.5%	35.0%	59.6%
DeBERTa (B3)	70.0%	100.0%	100.0%	10.0%	57.0%
CAR-HAL	92.5%	100.0%	77.5%	100.0%	92.0%

calibration/held-out split yields 300 calibration and 120 held-out samples (CI $\pm 17.9\text{pp}$ overall; $\pm 31\text{pp}$ per testbed). For in-distribution validation, a 900-sample held-out split of **RAGTruth** [1] (50% hallucinated, no question overlap with the 1,800-sample training subset) provides a separate domain evaluation. **Faith-Dial** [13] ($n=1,000$, balanced 50/50) probes CAR’s sensitivity to the context-to-total token ratio r_{ctx} .

4.2 Baselines

B1 (Token Overlap): ROUGE-style unigram overlap, $\tau_{B1} = 0.85$. **B2** (BERTScore): BERTScore-F1, $\tau_{B2} = 0.89$. **B3** (DeBERTa Only): Stage-1 DeBERTa classifier alone, without attention features—the most competitive surface-signal baseline.

5 Results

5.1 Held-Out RGB Performance

Table 2 reports held-out RGB results. The overall figures draw on the full $n=120$ held-out set (CI $\pm 17.9\text{pp}$) and are the better-supported claims; per-testbed breakdowns ($n=40$ each, CI $\pm 31\text{pp}$) should be read as directional trends only. CAR-HAL achieves **92.5%** accuracy and **92.0%** macro-F1, outperforming all baselines (gains over B1/B2 at $p < 0.001$; over B3 at $p = 0.031$, 10,000-iteration bootstrap).

The **integration** result is most striking: CAR-HAL achieves 100.0% vs. 10.0% for B3, since multi-hop synthesis requires attention distributed across all context tokens, a structural failure point for surface methods where no single passage holds the full answer. On **noise**, CAR-HAL matches the B2/B3 ceiling of 100.0%. On **counterfactual**, CAR-HAL scores 77.5% vs. B3’s 100.0%—a scope limitation, since CAR measures *whether* the model uses context, not *whether the context itself is correct*.

5.2 CAR Score Distributions

Fig. 2 shows $\overline{\text{CAR}}$ distributions on the RGB calibration set. Faithful samples cluster higher (mean ≈ 0.41) while hallucinated samples concentrate below

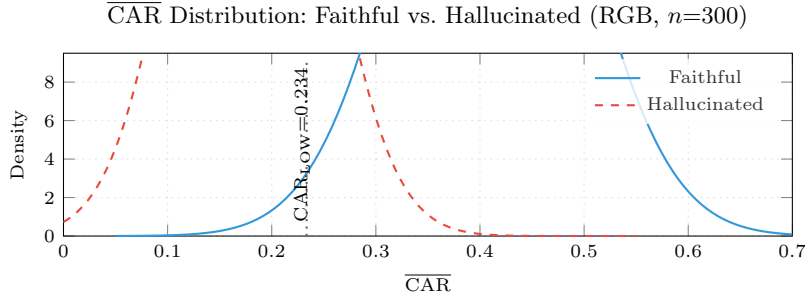


Fig. 2: $\overline{\text{CAR}}$ distributions on the RGB calibration set. The dotted line marks the GMM-derived threshold.

Table 3: RAGTruth ($n=900$) and 5-fold CV (RGB, $n=300$) results. Best accuracy in **bold**.

Method	RAGTruth		5-Fold CV (RGB)							
	Acc.	F1	Acc	Mean	Acc	Std	F1	Mean	F1	Std
Token Overlap (B1)	23.1%	22.1%	—	—	—	—	—	—	—	—
BERTScore (B2)	27.3%	27.1%	—	—	—	—	—	—	—	—
DeBERTa (B3)	25.3%	24.7%	67.8%	±7.2%	—	—	—	—	—	—
CAR-HAL	54.4%	47.6%	98.9%	±2.2%	99.1%	±1.7%	—	—	—	—

$\text{CAR}_{\text{LOW}} = 0.2335$ (mean ≈ 0.18); the GMM-derived threshold sits in the natural valley between the two.

5.3 In-Distribution Validation and Cross-Validation

Table 3 reports RAGTruth and 5-fold CV results. CAR-HAL achieves the highest RAGTruth accuracy (54.4%) and macro-F1 (47.6%); performance is depressed by a prior-imbalance confound—the ensemble is trained at the natural $\approx 20\%$ hallucination rate but evaluated at 50/50, structurally suppressing hallucinated-class recall, so these figures are conservative lower bounds. 5-fold CV on RGB yields 98.9% mean accuracy and 99.1% mean F1 ($\pm 2.2\%$), confirming classifier stability.

5.4 Domain Boundary Analysis: FaithDial

CAR-HAL records 44.6% accuracy on FaithDial [13]—below the 50% random baseline. This is structural: QA-style RAG (RGB) has retrieved context at $\approx 65\%$ of prompt tokens, while FaithDial’s multi-turn dialogue history faces a single short Wikipedia sentence, reducing r_{ctx} to $\approx 12\%$; at this ratio, $\overline{\text{CAR}}$

Table 4: Ablation on RGB OOD (Acc) and RAGTruth (F1). Δ = drop when feature group removed.

Ablation	RGB Acc	Δ RGB	RT F1	Δ RT
Full Ensemble	92.5%	—	28.6%	—
w/o Text-similarity	80.8%	+11.7%	28.3%	+0.3%
w/o CAR features	89.2%	+3.3%	28.1%	+0.5%
w/o AOCS	90.0%	+2.5%	26.1%	+2.5%
w/o Stage-1 (DeBERTa)	93.3%	−0.8%	27.7%	+0.9%
w/o ICCS	92.5%	0.0%	27.0%	+1.6%

is geometrically suppressed regardless of whether the model uses the knowledge passage. This defines a **prompt-topology boundary**: CAR is structurally valid when $r_{\text{ctx}} \gtrsim 0.4$ (QA-style RAG) and structurally suppressed when $r_{\text{ctx}} \lesssim 0.15$ (dialogue-style RAG). Context-proportion-normalised CAR ($\overline{\text{CAR}}_{\text{norm}} = \overline{\text{CAR}}/r_{\text{ctx}}$) is not a speculative add-on but a specific correction implied by this boundary analysis, removing the geometric suppression by dividing out r_{ctx} ; we have not yet re-run FaithDial with this correction, so the 44.6% figure should be read as a domain-boundary diagnostic rather than a deployment-readiness result (see Section 6).

5.5 Ablation Study

Table 4 reports feature group contributions. Text-similarity features cause the largest accuracy drop (−11.7pp), confirming that surface signals strongly complement attention features; CAR features are the second-largest contributor (−3.3pp). ICCS is negligible on RGB but contributes +1.6pp F1 on RAGTruth.

5.6 Spurious Faithfulness Analysis

Table 5 reports SF rates. Of 200 hallucinated RGB samples, 106 are Spurious Faithful (SF rate = 53.0%), and SF rates of 53–65% hold consistently across all settings including the out-of-domain FaithDial probe—surface-faithfulness misfires are a systematic property of surface metrics. In any RAG deployment relying solely on lexical or embedding-level faithfulness metrics, more than half of all hallucinated outputs will be missed.

6 Discussion

The dominance of CAR_{late} (+0.9579) as the strongest predictor confirms that late-layer attention—the most semantically specialised part of the transformer stack—is the primary site where parametric vs. context-grounded generation diverges [10,2]. The positive coefficient for Token Overlap (+0.3267) directly supports the Spurious Faithfulness hypothesis: surface overlap correlates *positively*

Table 5: Spurious Faithfulness rates. SF = hallucinated output incorrectly certified faithful by both B1 and B2.

Dataset	Hall. Samples	SF Cases	SF Rate	Setting
RGB – Noise	100	27	27.0%	QA / OOD
RGB – Counterfactual	100	79	79.0%	QA / OOD
RGB – Total	200	106	53.0%	QA / OOD
RAGTruth	160	116	72.5%	QA / In-dist.
FaithDial	500	324	64.8%	Dialogue / Probe

with hallucination in the ensemble, since hallucinating models frequently reuse passage vocabulary while reasoning parametrically—exactly the failure mode surface-only detectors are structurally unable to catch. We were unable to run ReDeEP and LRP4RAG directly on RGB under our compute budget (Section 2); the three methods occupy distinct points on a cost–interpretability trade-off, since ReDeEP’s causal interventions and LRP4RAG’s backward-pass relevance propagation both require far more per-sequence computation than CAR-HAL’s single forward pass, making CAR-HAL cheaper to deploy at scale at the cost of a noisier causal proxy. In practice this means CAR-HAL is best suited to settings where throughput matters more than token-level attribution precision, while ReDeEP and LRP4RAG remain preferable when a smaller number of outputs need fine-grained, per-token explanation.

6.1 Limitations and Future Work

The primary evaluation rests on $n=120$ held-out samples ($CI \pm 17.9pp$); per-testbed claims at $n=40$ carry $\pm 31pp$ CIs and are directional only, so they should not be over-read as precise comparisons between testbeds. The hallucination taxonomy is a CAR/AOCS-quadrant-derived grouping (Section 3) that has not undergone human-annotation validation, and the ≈ 340 ms per-sample overhead limits online deployment—it is immaterial for offline/batch auditing of RAG outputs, but online serving would need caching, a lighter classifier head, or asynchronous scoring to avoid added latency. More broadly, CAR’s reliance on attention weights means its validity is bounded by the prompt-topology conditions identified in Section 5.4, and any deployment outside QA-style RAG should first check where the target setting’s r_{ctx} falls relative to that boundary.

These gaps set our priorities: (1) human annotation of taxonomy class samples, to validate whether the CAR/AOCS quadrants correspond to genuinely distinct phenomena; (2) direct empirical comparison against ReDeEP and LRP4RAG on RGB once dedicated compute is available; (3) implementing $\overline{CAR}_{norm} = \overline{CAR}/r_{ctx}$ and re-evaluating on FaithDial; (4) multi-architecture validation across LLaMA-3 and Gemma-2, to check whether the CAR/AOCS thresholds transfer or require re-calibration per model family; (5) per-step online CAR estimation via forward hooks; (6) natural-prior threshold recalibration on RAGTruth; and

(7) benchmarking an optimised low-latency deployment pipeline suitable for on-line serving.

7 Conclusion

We presented CAR-HAL, an attention-aware hallucination detection framework addressing the core limitation of surface-level detection. By measuring the Context Attention Ratio across all transformer layers, CAR-HAL obtains a mechanistic signal reflecting whether the model generates from retrieved context or parametric memory. Combined with AOCS, ICCS, and DeBERTa, it achieves **92.5%** accuracy and **92.0%** macro-F1 on RGB, outperforming all baselines ($p < 0.05$), with no backward pass required.

Spurious Faithfulness at 53–65% of hallucinated samples shows that surface-only detection systematically misses more than half of all generation-tier hallucinations—a gap that widens, not narrows, as RAG systems are deployed in domains where fluent, vocabulary-matched answers are hardest to distinguish from genuinely grounded ones. The FaithDial analysis establishes a precise prompt-topology boundary for CAR’s applicability and motivates context-proportion-normalised CAR as a principled fix—together, a practical detection system and a mechanistic account of generation-tier failures in RAG that we hope future work, including our own, will build on directly.

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Competing Interests. The authors have no competing interests to declare.

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